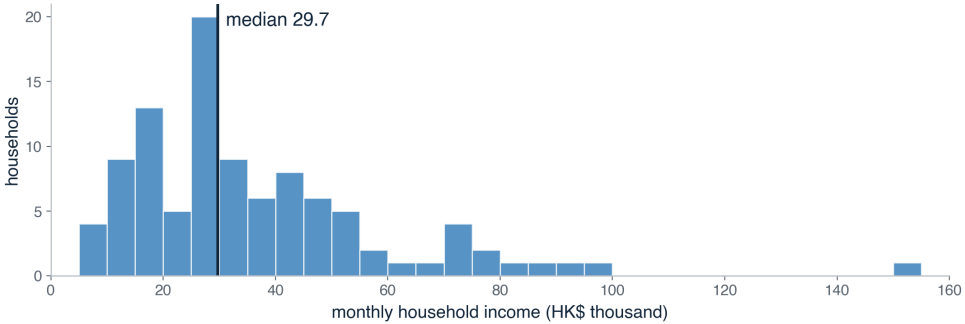
The Bootstrap

MATH 4432 Statistical Machine Learning · Tutorial T08

Department of Mathematics, HKUST

Fall 2026

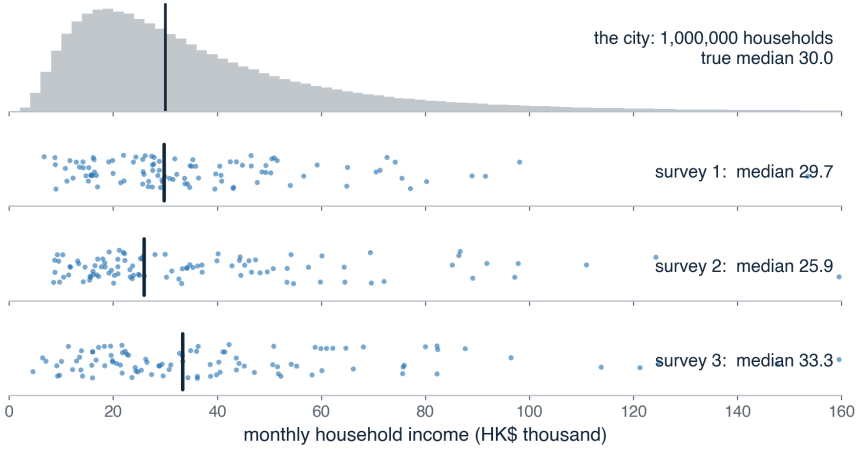
How sure are we?



a survey of 100 households: median income HK$29.7k

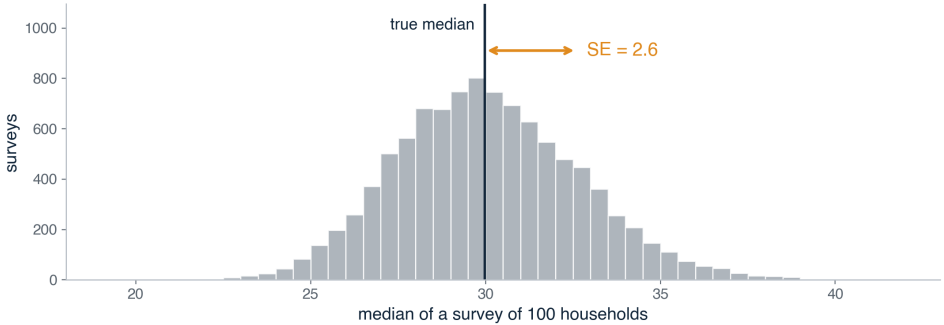
another 100 households would give a different number. How different?

Different samples, different estimates



a simulated city, so we know the truth

The sampling distribution



- **sampling distribution**: the values an estimate $\hat{\theta}$ takes over repeated samples
- **standard error** SE($\hat{\theta}$): its standard deviation here 2.6

One sample, no formula

with a formula

the mean:

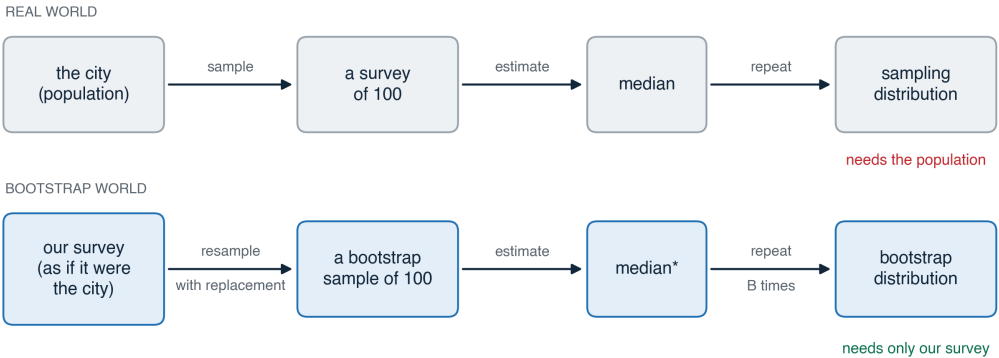
$$\text{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$$

without a simple formula

the median, a ratio, a correlation,
a coefficient when the model is wrong,
a prediction from a complex model, ...

we need the sampling distribution, but we have **one** sample

The bootstrap idea



treat our sample as if it were the population (the plug-in principle)

Resampling with replacement



- same size n as our sample; some observations twice, some not at all
- each observation is left out with probability $(1 - 1/n)^n \approx 0.37$

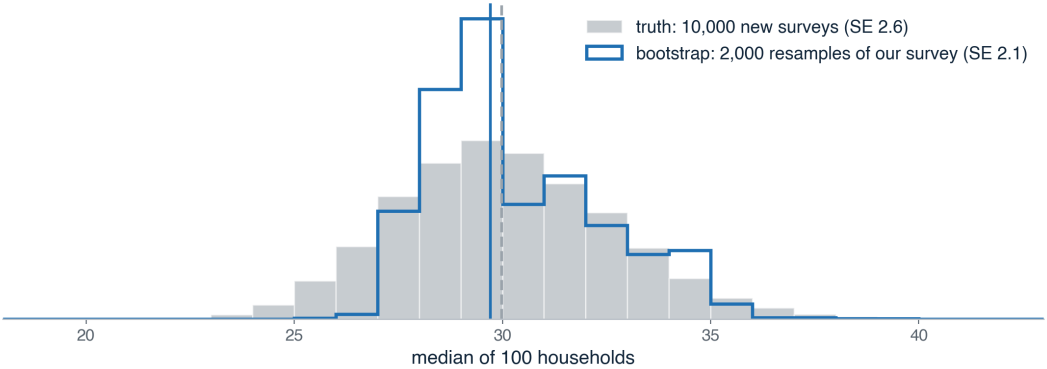
The bootstrap algorithm

1. draw B bootstrap samples of size n, with replacement (B = 1000 to 2000)
2. compute the estimate on each: $\hat{\theta}^*_1, \ldots, \hat{\theta}^*_B$
3. standard error and 95% interval from their spread:

$$\text{SE}_{\text{boot}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\theta}^*_b - \bar{\theta}^*\right)^2} \qquad 95\%\ \text{interval} = \left[\hat{\theta}^*_{(2.5\%)},\ \hat{\theta}^*_{(97.5\%)}\right]$$

$\bar{\theta}^*$: mean of the B bootstrap estimates · Efron (1979)

The bootstrap vs the truth



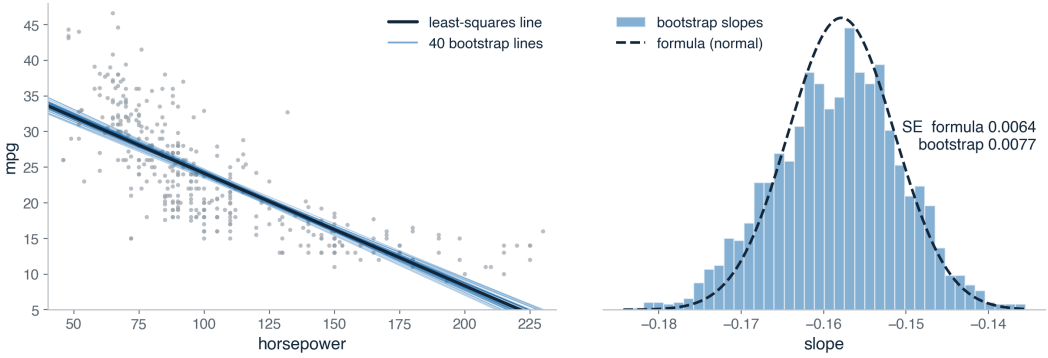
roughly the right spread from one survey · centred at our median, not at the truth

Cross-validation and the bootstrap

	cross-validation (T07)	**bootstrap**
question	how well does a model predict?	how uncertain is an estimate?
resampling	split into folds, no repeats	draw n with replacement
output	test error	standard error, confidence interval
used later in	model selection	bagging, random forests

both reuse the one sample we have · ISL chapter 5

Auto: mpg and horsepower

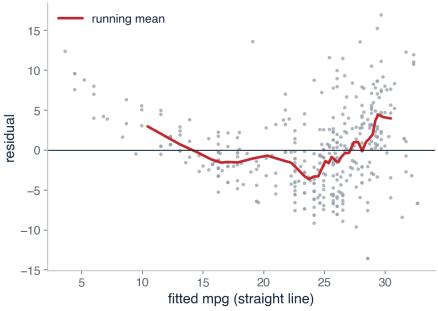


392 cars · resample cars, refit the line, 2,000 times

Formula or bootstrap?

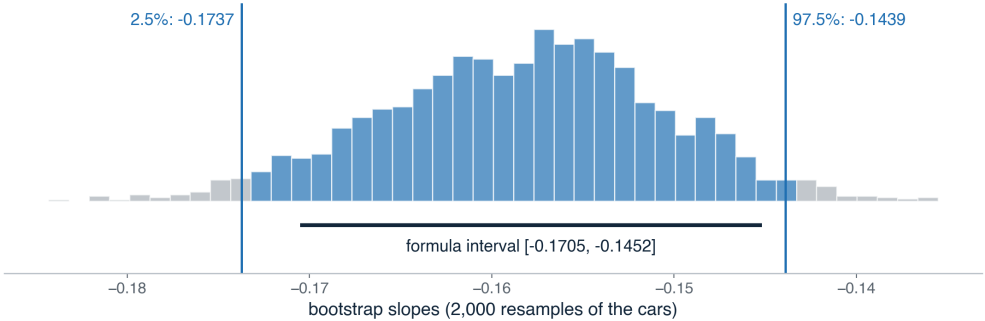
		standard error	
model	coefficient	formula	bootstrap
line	intercept	0.717	0.881
	horsepower	0.00645	0.00767
quadratic	intercept	1.80	2.10
	horsepower	0.0311	0.0336
	horsepower^2	0.000122	0.000122

the formula assumes the model is right and the noise has constant variance



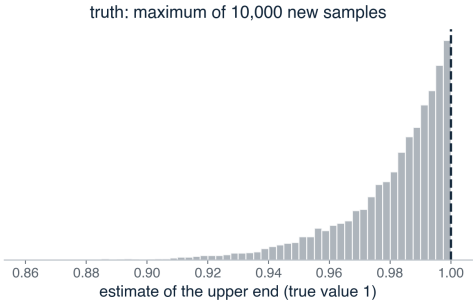
residuals: a curve and a funnel

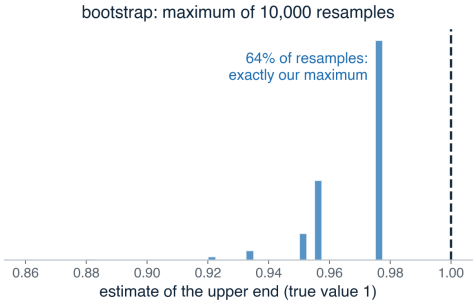
Bootstrap confidence intervals



- **percentile interval**: the middle 95% of the bootstrap estimates
- 95%: over repeated studies, about 95% of such intervals contain the true value

When the bootstrap fails





estimating the largest possible value: a resample never exceeds our largest observation

- extremes · very small samples · dependent data (time series)

1 · the bootstrap in a few lines

```
def bootstrap(data, stat, B=2000, seed=0):
    rng = np.random.default_rng(seed)
    idx = rng.integers(0, len(data), size=(B, len(data)))   # B samples, with replacement
    return np.array([stat(data[i]) for i in idx])

boot = bootstrap(incomes, np.median)
print(np.median(incomes).round(1), boot.std(ddof=1).round(2),
      np.percentile(boot, [2.5, 97.5]).round(1))
boot_mean = bootstrap(incomes, np.mean)
print(boot_mean.std(ddof=1).round(2),
      (incomes.std(ddof=1) / np.sqrt(len(incomes))).round(2))
```

```
29.7 2.07 [27.5 34.8]     # median, its bootstrap SE, 95% interval
2.32 2.34                 # mean: bootstrap SE, formula s/sqrt(n)
```

2 · bootstrap a regression

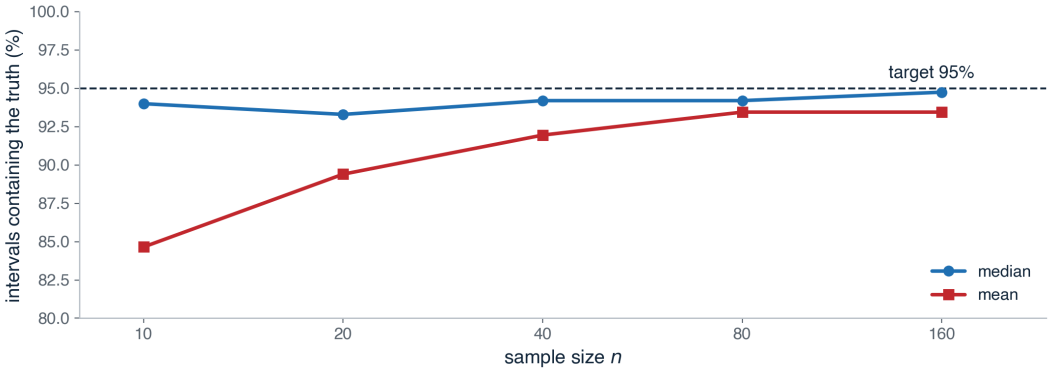
```
auto = pd.read_csv("data/Auto.csv", na_values="?").dropna()
Xy = auto[["horsepower", "mpg"]].to_numpy(dtype=float)
slope = lambda d: np.polyfit(d[:, 0], d[:, 1], deg=1)[0]       # refit on resampled cars
boot = bootstrap(Xy, slope)

fit = sm.OLS(auto.mpg, sm.add_constant(auto.horsepower)).fit()
print(fit.bse["horsepower"].round(4), boot.std(ddof=1).round(4))
print(fit.conf_int().loc["horsepower"].values.round(4),
      np.percentile(boot, [2.5, 97.5]).round(4))
```

```
0.0064 0.0077                          # SE: formula, bootstrap
[-0.1705 -0.1452] [-0.1737 -0.1439]    # 95% CI: formula, bootstrap
```

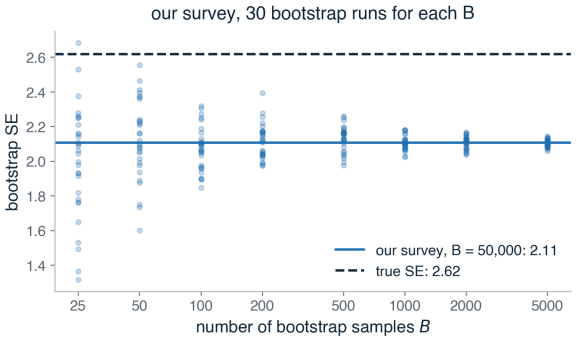
resample whole rows: each car keeps its own horsepower and mpg

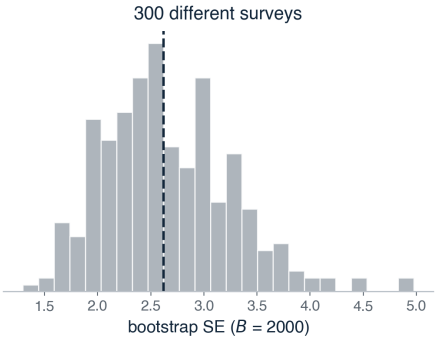
3 · does a 95% interval cover 95%?



skewed data and small n: the interval for the mean is too short

4 · how many bootstrap samples?





larger B: less simulation noise, same sampling noise

Take away

- an estimate changes from sample to sample; its **standard error** measures how much
- **bootstrap**: resample our data with replacement, recompute, look at the spread
- works for almost any estimate, without a formula or a correct model
- larger B removes simulation noise only; one sample is still one sample
- it can fail: extremes, very small samples, dependent data
- cross-validation → prediction error; bootstrap → uncertainty

References

- James, Witten, Hastie & Tibshirani, *An Introduction to Statistical Learning*, §5.2 and lab §5.3.4 — the Auto example
- Efron (1979), Bootstrap methods: another look at the jackknife, *Annals of Statistics*
- Efron & Tibshirani (1993), *An Introduction to the Bootstrap*
- Hastie, Tibshirani & Friedman, *The Elements of Statistical Learning*, §7.11, §8.2
- `T08_Bootstrap.ipynb` — every number on these slides